

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2456763.72346 +/- 0.00073 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.113 +/- 0.023
Transit depth (Rp/Rs)^2	1.29 +/- 0.51 [%]
Orbital Inclination (inc)	87.06 +/- 2.38
Airmass coefficient 1 (a1)	1.177 +/- 0.014
Airmass coefficient 2 (a2)	-0.107 +/- 0.011
Scatter in the residuals of the lightcurve fit is	1.23 %
Best Comparison Star	None
Optimal Aperture	3.52
Optimal Annulus	12.0
Transit Duration (day)	0.057 +/- 0.0033

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.113 $\pm$ 0.023 vs 0.1594 (deviation 2.02 σ)
Duration fit vs published	0.057 d vs 0.0512 d (deviation 1.74 σ)
Mid-time O-C	1.6 min
Depth SNR	4.9
Residual scatter	1.23 %

Light curve

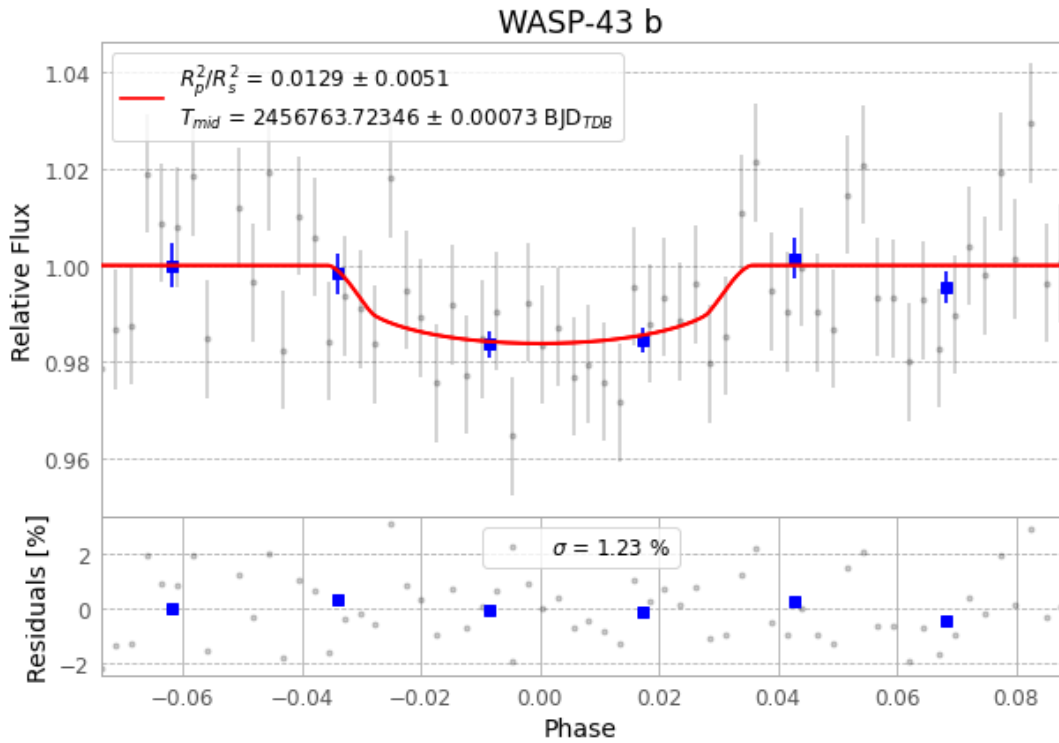


Figure 1 — FinalLightCurve_WASP-43 b_2014-04-15.png (SHA-256 076586da3222f9c3...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [242, 223], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 53e863cf996f849f...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

64 FITS frames (42 MB), WASP-43140416034812.FITS → WASP-43140416065712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-140416.log (a8edce4eaa6a...), FinalLightCurve_WASP-43 b_2014-04-15.png (076586da3222...), FinalLightCurve_WASP-43 b_2014-04-15.csv (0bb1e93415af...)

Compute resources

Wall time 117.9 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-25 00:24Z.